# CGA Math

Spencer T. Parkin

July 11, 2026

This paper documents the math needed to compose and decompose elements of CGA.

## Foundation

For any blade $B$ of CGA, it represents two geometries. These are all vectors $x$ in the euclidean sub-algebra of CGA such that

$$p(x) \cdot B = 0,$$

or all such vectors with

$$p(x) \wedge B = 0,$$

where $p(x)$ is defined as the vector

$$p(x) = o + x + \frac{1}{2}x^2\infty.$$

We'll call the first of these the "dot representation" and the second, "wedge representation". To convert from one to the other, you can multiply by $I$, the unit psuedo-scalar. This is because

$$p(x) \cdot B = 0 \iff p(x) \wedge BI = 0.$$

Of course, $B$ and $BI$ are duals of one another, so the conversion goes both ways.

Using blades to represent geometries, we get the following useful properties. If $B_1$ and $B_2$ are blades with $B_1 \wedge B_2 \neq 0$, then we have

$$\{x|p(x) \cdot B_1 \wedge B_2 = 0\} = \{x|p(x) \cdot B_1 = 0\} \cap \{x|p(x) \cdot B_2 = 0\},$$

as well as

$$\{x|p(x) \wedge B_1 \wedge B_2 = 0\} \supseteq \{x|p(x) \wedge B_1 = 0\} \cup \{x|p(x) \wedge B_2 = 0\}.$$

In this latter case, since there are only so many shapes in CGA, we can reasonably deduce that $B_1 \wedge B_2$ will be the geometry that fits those represented by $B_1$ and $B_2$. You may, for example, find the sphere that fits a given circle and point, or the circle that fits 3 given points, and so on.

The shapes of CGA are either round or flat. If using the wedge presentation, a round can be converted into a flat by tacking on $\infty$ with the outer product. For example, if $B$ represents a circle using the wedge representation, then $B \wedge \infty$ is the plane containing the circle, also in the wedge representation.

# Compositions

For all geometry that follows, we let $c$ and $n$ denote euclidean vectors, and $r$ denote a scalar. We always require that $|n| = 1$.

We are going to use the dot representation here instead of the wedge representation, and ignore the magnitude of $B$ as long as it is non-zero.

## Planes

Letting $B = n + (n \cdot c)\infty$, we have

$$\begin{aligned} 0 &= p(x) \cdot B \\ &= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot (n + (c \cdot n)\infty) \\ &= -c \cdot n + x \cdot n \\ &= (x - c) \cdot n, \end{aligned}$$

which is clearly the equation for a plane centered at $c$ with normal $n$.

## Lines

Lines are a bit trickier. Here, let $i = e_1e_2e_3$ denote the unit psuedo-scalar of the 3D euclidean sub-algebra of CGA. Doing so, let $B = ni + (ni \cdot c)\infty$. We then see that

$$
\begin{aligned}
0 &= p(x) \cdot B \\
&= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot (n - (c \cdot ni)\infty) \\
&= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot (n + \infty \wedge (c \cdot ni)) \\
&= (o \cdot \infty)(c \cdot ni) + x \cdot ni - (x \cdot (c \cdot ni))\infty \\
&= -c \cdot ni + x \cdot ni - (x \cdot (c \wedge n)i)\infty \\
&= (x - c) \cdot ni - (x \wedge c \wedge n)i\infty.
\end{aligned}
$$

Here, we must have $0 = (x - c) \cdot ni$ as well as $0 = x \wedge c \wedge n$. The first of these is clearly a line centered at $c$ with normal $n$, which is what we're after, but what of the second? Well, we need only consider those values of $x$ for which the first is satisfied. If $x = c$, we have a trivial case, and all is well. Otherwise, we have $x - c = \lambda n$ for some non-zero scalar $\lambda$, which means that $x \wedge c \wedge n = x \wedge c \wedge \frac{1}{\lambda}(x - c)$, which is clearly zero.

## Spheres

Consider $B = o + c + \frac{1}{2}(c^2 - r^2)\infty$. We then have

$$
\begin{aligned}
0 &= p(x) \cdot B \\
&= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot \left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right) \\
&= -\frac{1}{2}(c^2 - r^2) + x \cdot c - \frac{1}{2}x^2.
\end{aligned}
$$

Rearranging this, we find that

$$
\begin{aligned}
r^2 &= c^2 - 2(x \cdot c) + x^2 \\
&= (x - c)^2,
\end{aligned}
$$

which is clearly the equation for a sphere.

Interestingly, we also find imaginary spheres in CGA with $B = o + c + \frac{1}{2}(c^2 + r^2)\infty$. Imaginary rounds are necessary in cases where an intersection is computed that can't actually exist.

## Points

Points are simply the special case of spheres with $r = 0$. A point at $c$ is simply $p(c)$. Note that these, technically, are round points, not to be confused with tangent points or flat points.

For a flat point, let

$$B = (n + (n \cdot c)\infty) \wedge (ni + (ni \cdot c)\infty),$$

which is clearly the intersection of a line and a plane. If you were to expand this wedge product out, things would cancel, and you'd be left with

$$B = i + ci \wedge \infty,$$

showing that indeed $n$ is not applicable here.

Note that we can recognize the relationship between round and flat points with

$$B = (p(c) \wedge \infty)I,$$

where $I$ is the unit psuedo-scalar of CGA. In general, tacking on $\infty$ to a round gives you the corresponding flat, as long as the round is in wedge representation.

## Circles

Here, let

$$B = (n + (n \cdot c)\infty) \wedge \left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right),$$

which is the intersection of a sphere centered on a plane. Use an imaginary sphere here to get an imaginary circle. Such circles form the intersection of, for example, two real spheres that are intersected, but that don't actually intersect.

## Point-Pairs

Point-pairs may seem odd, but they're simply the 1-dimensional analog of spheres and circles. Here, let

$$B = (ni + (ni \cdot c)\infty) \wedge \left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right).$$

This is simply the intersection of a line going through the center of a sphere. Similar to cricles, use an imaginary sphere here to get an imaginary point-pair. Such a point-pair would occur, for example, when you compute the intersection of a line and a sphere that don't actually intersect.

### Tangent Points

These are the special case of circles or point-pairs when $r = 0$. Note that while it takes 4 points to fit a sphere, it only takes 2 tangent points, or 2 points and 1 tangent point, to do the same. You can get a tangent point by intersecting two spheres that touch in a single point.

## Decomposition

Here we utilize the composition forumlas derived earlier in order to extract the variables $c$, $n$ and $r$ from a given blade $B$. Since we can't ignore magnitude here, we need an extra scalar $w$ to represent the "weight" of the geometry.

It should be noted here that given a blade $B$, even if we know how to decmpose any shape, exactly what shape to decompose $B$ as is not immediately clear. One idea is to choose all that apply for the grade. If a decomposition method succeeds, it will have been correct. All others should fail with division by zero or something similar.

### Planes

Let $B = w(n + (n \cdot c)\infty)$. We then have

$$o \cdot B \wedge \infty = wn.$$

We can then normalize this to recover $n$ and use the length to recover $w$. Note that in a computer program, we might as well skip this and just pick out the $wn$ term from the vector. In any case, we can recover $c$ by letting $c = \lambda n$ for some scalar $\lambda$, and then see that

$$\frac{o \cdot B}{-w} = n \cdot c = \lambda.$$

Likewise, in a computer program, we could just extract $w\lambda$ by picking out the coeficient of $\infty$.

## Lines

Let $B = w(ni + (ni \cdot c)\infty)$. We then have

$$-(o \cdot B \wedge \infty)i = wn.$$

Here we can recover $w$ then as the length, and normalize to recover $n$. To find $c$, let $c \cdot n = 0$, and write

$$\left(\frac{o \cdot B}{w}\right) i = c \wedge n = cn.$$

Multiplying this on the right by $n$ in the geometric product then gives us $c$. For a computer program, we could just find $wc$ directly by examining the term with $\infty$.

## Spheres

Let $B = w\left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right)$. Then

$$-\infty \cdot B = w,$$

and

$$\frac{o \cdot B \wedge \infty}{w} - o = c,$$

but, of course, in a computer program, just extract $wc$ directly from the vector. We also have

$$r^2 = c^2 + \frac{2}{w} o \cdot B.$$

Here we can determine if the sphere is real or imaginary by the sign of the right-hand side. Then just calculate $r$ by taking the square root of the absolute value of the right-hand side.

## Points

The decomposition of a round point is the same for that of a sphere except that we expect the radius to be zero.

As for flat points, let $B = w(i + ci \wedge \infty)$. We can easily pick off $w$ as the multiple of $i$ term, but a formula can be given by

$$-(o \cdot B \wedge \infty)i = w.$$

Similarly, we can get

$$\left(\frac{o \cdot B}{w}\right) i = c,$$

but again, a computer program should just pick out the term directly.

## Circles

When we start thinking about decomposing blades of grade greater than one, one idea is to factor first, then try decomposition on the vector factors. But factoring is not a trivial matter, and I'm not sure if just any factorization would work.

Let $B = w(n + (n \cdot c)\infty) \wedge \left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right)$ and expand to get

$$B = wn \wedge o + wn \wedge c + \frac{w}{2}(c^2 - r^2)n \wedge \infty + w(n \cdot c)\infty \wedge o + w(n \cdot c)\infty \wedge c.$$

A computer program could pick out $wn$ directly, or we could calculate it as

$$o \wedge \infty \cdot B \wedge \infty = wn.$$

We can now recover $w$ and $n$. We recover $c$ with

$$c = n\left[\frac{\infty \wedge o}{w} \cdot (B \wedge \infty \wedge o + B)\right].$$

Again, a computer program can pick out the terms easier to reduce the number of calculations needed. For $r^2$, we find that

$$r^2 = c^2 - 2(c \cdot n)^2 - n \cdot \frac{2}{w}\left[\infty \wedge o \cdot B \wedge o\right].$$

Here again we can determine if the circle is real or imaginary and then compute $r$.

## Point-Pairs

Let $B = w(ni + (ni \cdot c)\infty) \wedge \left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right)$ and expand to get

$$B = wni \wedge o + wni \wedge c + \frac{w}{2}(c^2 - r^2)ni \wedge \infty + w(ni \cdot c) \wedge \infty \wedge o + w(ni \cdot c) \wedge \infty \wedge c.$$

We then have

$$-\left(o \wedge \infty \cdot B \wedge \infty\right) i = wn$$

from which $w$ and $n$ may be recovered. We recover $c$ with

$$c = -n\left[\frac{\infty \wedge o}{w} \cdot (B \wedge \infty \wedge o + B)\right] i.$$

For $r^2$ we find that

$$r^2 = -c^2 + 2(c \cdot n)^2 + n \cdot \frac{2}{w}\left[\infty \wedge o \cdot B \wedge o\right] i,$$

from which we can find $r$ and determine if the point-pair is real or imaginary.

### Tangent Points

Write this...

## Transformations

Using the wedge representation, it's not hard to show that any of the conformal transformpations can be applied to any CGA shape. The idea is simply that if $V$ is a versor, and $B_k = b_1 \wedge \cdots \wedge b_k$ is a $k$-blade, then

$$VB_kV^{-1} = \bigwedge_{i=1}^{k} Vb_kV^{-1} = \bigwedge_{i=1}^{k} Vp(x_k)V^{-1},$$

where the points $x_k$ are suitabily chosen. This shows that a set of points used to fit the shape are transformed by the versor into some other set of points that will fit some other shape. Thus, we only need to understand how the versor transforms points and $\infty$. This is analogous to any linear transformation reducing to an understanding of how the axes are transformed. Here, for each transformation, we need only consider how it transforms $p(x)$.

But what about the dot representation? Well, if we write

$$VB_kV^{-1} = -VB_kI^2V^{-1} = \pm VB_kIV^{-1}I,$$

then we see that they transform the same way.

So what are the transformations? Since versors are geometric products of vectors, we first look at what vectors do. All other transformations are composed of a sequence of what transformations vectors can perform. What we find is that vectors can perform reflections of inversions. Interestingly, all rigid body transformations come from just reflections.

## Reflections

If we let $V$ be the dot representation of a plane, then $Vp(c)\tilde{V}$ reflects $c$ about the plane. If $x$ is the smallest vector such that $c + x$ is on the plane, then the reflected point is $c + 2x$.

## Inversions

If we let $V$ be the dot representation of a sphere, then $Vp(c)\tilde{V}$ inverts $c$ about the sphere. But what in the world does this mean? Is the point reflected across the surface of the sphere? That was my initial thought, but to really know what's going on, we need to do the math.

If we let $V = o - \frac{1}{2}r^2\infty$ be the dot representation of a sphere centered at origin with radius $r$, then

$$\frac{Vp(c)\tilde{V}}{-c^2} = p(r^2c^{-1}).$$

Here we see where the transform gets its name. Seeing where the point is going to land in all cases is still not terribly intuitive with this result. Also, what if we move the sphere away from origin? Does the problem just translate? I'm going to leave these questions open for now and move on.

## Rotations

These are just double reflections in distinct planes intersecting in the axis of rotation. The angle between the planes is half the desired angle of rotation.

## Translations

These are just double reflections in distinct yet parallel planes. The distance between the planes is half the desired distance of translation.

## Uniform Scale

These are also called diolations. What we do here is a double inversion about concentric spheres. If

$$V = (o + \frac{1}{2}(-r_b^2)\infty)(o + \frac{1}{2}(-r_a^2)\infty),$$

then

$$\frac{Vp(c)\tilde{V}}{r_a^4} = p\left(\frac{r_b^2}{r_a^2}c\right).$$

Therefore, letting $r_a = 1$ arbitrarily, we can determine $r_b$ from a given scale factor $s$ as $r_b = \sqrt{s}$.

## Transversions

Here we're going to mix reflections with inversions.

Do some research and talk more about it here...

# Appendix

## Helpful Identities

Let $a$ be a vector and $A_k$ be a $k$-blade. We then have

$$\begin{aligned} a \wedge A_k &= \frac{1}{2}\left(aA_k + (-1)^k A_k a\right), \\ a \cdot A_k &= \frac{1}{2}\left(aA_k - (-1)^k A_k a\right). \end{aligned}$$

Then, for euclidean vectors $a$ and $b$, this helps us find

$$\begin{aligned} (a \cdot b)i &= a \wedge bi, \\ (a \wedge b)i &= a \cdot bi. \end{aligned}$$

Similar identities are often useful when we want to pull $i$ in or out.

You should also know

$$a \cdot A_k = -\sum_{i=1}^{k} (-1)^i (a_1 \wedge \cdots \wedge a_{i-1} \wedge a_{i+1} \wedge \cdots \wedge a_k),$$

where $A_k = \bigwedge_{i=1}^k a_k$.